

Digital Transformation and Firm Productivity: A Synthetic Workflow Demo

Economics Research Workflow Workshop

2026-06-06

Abstract

This short draft demonstrates how a VS Code centered workflow can connect simulated firm-level data, Stata DID estimation, Matlab theory calibration, Jupyter notebooks, and formal L^AT_EX writing. The data are synthetic and should not be interpreted as evidence about real firms or policies.

1 Research Question

Digital transformation may raise firm productivity by improving information flows, production scheduling, inventory control, and managerial decision-making. The empirical question in this demo is whether firms that adopt digital technologies experience higher total factor productivity after adoption, relative to comparable firms that have not yet adopted.

2 Empirical Specification

The synthetic panel follows firms from 2016 to 2024. Treated firms adopt digital technologies beginning in 2020. The baseline DID specification is

$$\log(TFP_{it}) = \alpha_i + \lambda_t + \tau Digital_{it} + X'_{it}\beta + \varepsilon_{it}, \quad (1)$$

where α_i denotes firm fixed effects, λ_t denotes year fixed effects, and X_{it} includes capital intensity, export share, and ownership controls. Standard errors are clustered at the firm level.

3 Identification Checks

The workflow produces a pre-treatment parallel-trend figure and a placebo test that assigns a fake treatment date in the pre-period. These diagnostics are not proof of identification, but they are useful first checks before interpreting the DID coefficient.

4 A Simple Theory Channel

The Matlab component formalizes a compact mechanism. A firm chooses digital intensity x and obtains productivity gain

$$g(x, m) = \phi x + \lambda m x - \frac{1}{2}\psi x^2 - c, \quad (2)$$

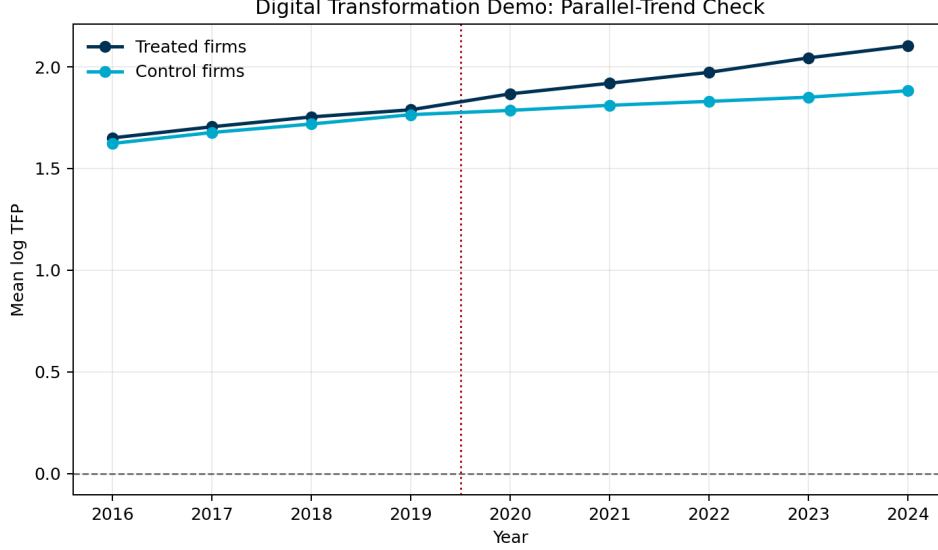


Figure 1: Parallel-trend check in synthetic firm panel data.

where m is managerial capability, ϕ is the baseline productivity return to digital adoption, ψ captures convex implementation costs, and c is a fixed adoption cost. The script calibrates ϕ so that the model-implied average gain is close to the simulated DID estimate.

5 Draft Result Summary

In the synthetic data, digital transformation is associated with higher log productivity after adoption. The event-study output is designed to show small pre-treatment differences and growing post-adoption gains. The placebo test checks whether a fake pre-period treatment generates a similar effect. Because all data are simulated, the estimate should be read only as a workflow demonstration, not as a substantive claim about real-world digital transformation.

6 Reproducibility Note

The analysis is generated from three executable components: a Python data-generation script, a Stata DID script, and a Matlab theory-calibration script. The Jupyter notebooks provide interactive debugging surfaces, while this $\text{\LaTeX}$ file records the formal writing layer.

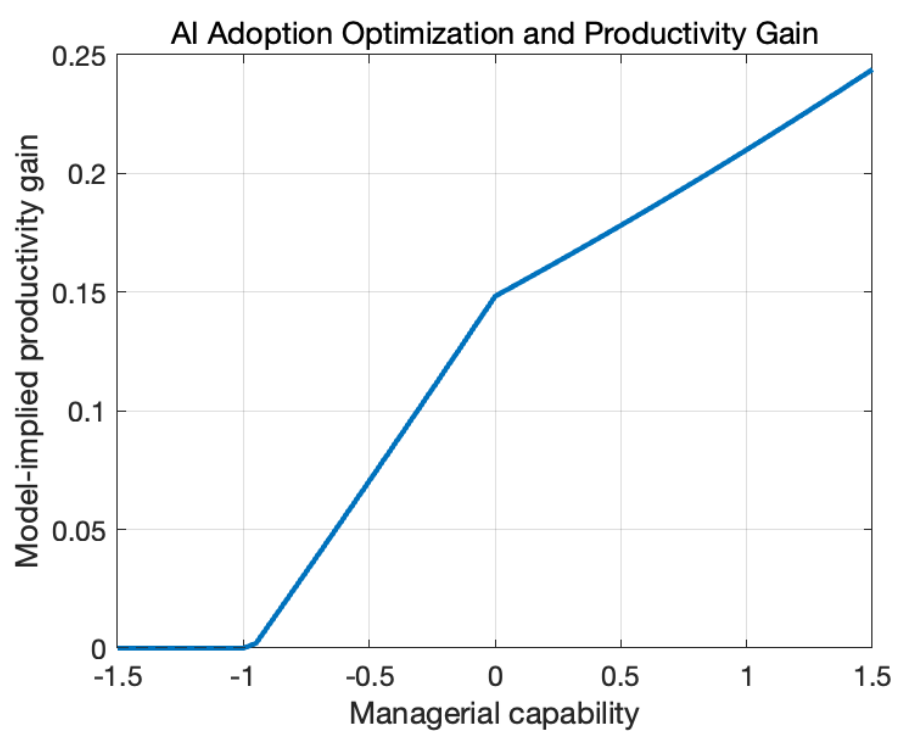


Figure 2: Model-implied productivity gain by managerial capability.